

variables in the participation equation. Thus

$$Y_1 = g_1(T) + U_1, \quad (7a)$$

$$Y_0 = g_0(T) + U_0, \quad (7b)$$

and $\Pr(D=1|Z)$ is the probability of programme participation. Since outcomes are measured after enrollment, and different factors operate on outcomes and enrollment decisions, Z and T may contain distinct variables, although they may share some variables in common. We consider several generalizations of (A-3) and (A-4) that apply to the residuals from models 1(a) and 1(b): $U_0 \perp\!\!\!\perp T|D, Z$ so

$$\Pr(U_0 \leq u_0 | T, Z, D) = \Pr(U_0 \leq u_0 | Z, D), \quad (A-3')$$

and $U_0 \perp\!\!\!\perp D|P(Z)$ or

$$E(U_0 | T, Z, D) = E(U_0 | Z, D), \quad (A-4')$$

and $E(U_0 | P(Z), D=1) = E(U_0 | P(Z), D=0)$.

Under these assumptions, it is possible to improve on the efficiency of the unrestricted matching estimator and invoke the exclusion restrictions in (A-4'). This leads to the *regression-adjusted matching estimator* formally justified in our companion paper (Heckman, Ichimura and Todd (1997)) and empirically implemented in this paper.

We extend matching to a panel or repeated cross-section context in a new nonparametric conditional difference-in-differences estimator. This estimator was first proposed in Heckman, Ichimura, Smith and Todd (1995a, revised 1996b), but has precedents in the work of Heckman and Robb (1985, 1986). Let t represent a time period after the programme start date and t' a time period before the programme. Our conditional difference-in-differences estimator compares the conditional before-after earnings of programme participants with those of non-participants. It extends the conventional difference-in-differences estimator by defining outcomes conditional on X and using semiparametric methods to construct the differences. The population moments to which the estimator converges under standard conditions are

$$D_{t,t'}(X) = E(Y_{1t} - Y_{0t} | X, D=1) - E(Y_{0t} - Y_{0t'} | X, D=0).$$

An estimator based on sample analogues to these population moments is robust to temporally-persistent separable components of bias including those that might arise from geographical or questionnaire mismatch between participants and members of the control group.

Term $D_{t,t'}(X)$ identifies $E(\Delta | X, D=1)$ if the following assumption holds

$$E(Y_{0t} - Y_{0t'} | X, D=1) = E(Y_{0t} - Y_{0t'} | X, D=0). \quad (A-5)$$

Under additive separability this condition is equivalent to

$$E(U_{0t} - U_{0t'} | X, D=1) = E(U_{0t} - U_{0t'} | X, D=0),$$

or $B_t(X) = B_{t'}(X)$. Under index sufficiency the condition becomes

$$E(U_{0t} - U_{0t'} | P(Z), D=1) = E(U_{0t} - U_{0t'} | P(Z), D=0). \quad (A-5')$$

If Assumptions (A-3) or (A-4) or (A-3') or (A-4') are satisfied at times t and t' , the difference-in-difference version of the estimator will also be justified. However Assumptions (A-5) and (A-5') are weaker than those assumptions and are consistent with the index-sufficient sample selection bias model, as noted in Heckman, Ichimura, Smith and Todd

(1996b). The difference-in-difference estimator is less demanding of the data than the sample selection estimator in that it does not require a set of X values where there is no selection bias (i.e. for which $E(U_0|X, D=0)=0$).¹² However, it is also more demanding because it requires pre-programme data. From an economic standpoint it is an attractive estimator because, unlike conventional matching estimators, it permits selection to be based on potential programme outcomes and allows for selection on unobservables. In particular, it is consistent with a Roy model of self selection applied to a panel setting. Heckman and Robb (1985) present a parametric version of our difference-in-differences estimator, which is used in Ashenfelter and Card (1985).

5. OUR PREVIOUS EVIDENCE ON THE FUNCTIONAL FORM OF SELECTION BIAS

In a series of papers (Heckman and Roselius (1993), Heckman, Ichimura, Smith and Todd (1994, revised 1996b), Todd (1995)), we have used semiparametric methods to characterize the form of the bias that arises from using mean outcomes of comparison group members to proxy the mean outcomes that participants in a programme would have experienced if they had not participated in it. Let $B(X)$ be the bias for a particular value of X . It is defined as

$$B(X) = E(Y_0|X, D=1) - E(Y_0|X, D=0). \quad (8)$$

Under the conditions that justify matching, $B(X)=0$ for all X where the bias is defined. For the additively-separable case of (1a) and (1b), the bias is

$$B(X) = E(U_0|X, D=1) - E(U_0|X, D=0). \quad (9)$$

Our papers present evidence for a variety of demographic groups that the bias function $B(X)$ is described by two main features. First, assuming additive separability, the bias has an *index property*. Let $P(X)=\Pr(D=1|X)$ be the probability of participation in the programme. If the bias has the index property

$$B(X) = \tilde{B}(P(X)), \quad (P-1)$$

where $\tilde{B}$ is a function of a single index, $P(X)$, and X enters the bias function solely through the index. This representation is consistent with a broad class of widely-used index function models described in Heckman and MaCurdy (1985) and is at the heart of the conventional sample selection bias model (Heckman (1980)) and the Roy model of self selection (Heckman (1990b) and Heckman and Honoré (1990)). An index representation greatly simplifies the characterization of the problem of evaluation bias and focuses attention on the probability of selection as a central ingredient to the formulation and solution of the evaluation problem. It plays an important role in the development of semiparametric methods for solving the selection problem, which we present elsewhere. (Heckman, Ichimura, Smith and Todd (1994, 1996b).) Observe that if $\tilde{B}(P(X))$ is the same in periods t and t' , then if $Z=X$, Assumption (A-5') is justified. Thus an estimator based on the conditional difference-in-differences moment condition (A-5') is consistent with the index-sufficient selection estimator if, for example, the bias is constant over time or if it is symmetric around the date of entry into the programme ($t=0$). Heckman, Ichimura, Smith and Todd (1996b) present graphical evidence in support of such symmetry except for low values of P for the data analysed in this paper.

12. Heckman (1990a, b) and Heckman, Ichimura, Smith and Todd (1996b) discuss the importance of this condition in using the index-sufficient self selection model.

Second, our research in this paper and other papers (Heckman, Ichimura, Smith and Todd (1996a, b)) investigates whether the bias is *balanced* both for participants and controls i.e. whether

$$B(X) = \tilde{B}(P(X)) = 0. \quad (\text{P-2})$$

This is the consequence of Assumptions (A-1) and (A-2) that are invoked to justify matching including the weaker version introduced in Heckman and Robb (1986). The research reported in this paper decisively rejects (P-2) for four demographic groups.

6. DETERMINANTS OF PROGRAMME PARTICIPATION

The probability of participation in the programme being evaluated ($P(X)$) is a key ingredient of our empirical strategy for characterizing and solving the evaluation problem. It is a central feature of the econometric model for index-sufficient selection bias and for matching (see Heckman and Robb (1986)). Heckman and Smith (1994) present an extensive analysis of the determinants of participation in the JTPA programme. We briefly summarize their findings concerning the relative importance of background characteristics, recent labour force status and earnings histories in the participation process for eligible persons.

Their main conclusion is that for all demographic groups recent unemployment histories are important predictors of participation in training programmes. Trainees enter the JTPA programme as a form of job search. For adult women, recent marital histories are also important since recently-divorced or separated women in life cycle transitions are more likely to participate in the programme than are others. Models of the participation decision based on variables that predict job-seeking are much better able to predict participation than are models that include only demographic characteristics like education, age and race.

Heckman and Smith find a dip in the earnings of participants shortly before they apply to the programme. This earnings pattern was first noted by Ashenfelter (1978) for a predecessor to the JTPA programme. However it is unemployment dynamics and not earnings dynamics that best predicts who goes into the programme. In terms of labour force status, unemployment peaks for participants just prior to the date of enrollment but there is little change in unemployment status for members of a nonexperimental comparison group. Unemployment increases for participants both because employed persons lose their jobs and seek the assistance of training programmes and because persons previously out of the labour force enter it and use training programmes as a vehicle for labour force entry. Ashenfelter's earnings dip is a consequence of a more fundamental process of unemployment dynamics. The evidence presented by Heckman and Smith shifts the emphasis in the evaluation of job training programmes away from a focus on controlling for earnings histories, and longitudinal methods based on them (as developed in Ashenfelter (1978), Heckman (1978), Heckman and Robb (1985, 1986), and Ashenfelter and Card (1985)), and toward models based on unemployment dynamics as predictors of participation and as variables to control for bias. They further note that the main determinants of participation are not merely the consequence of eligibility rules. Among eligible persons, persons seeking work are more likely to enter job training programmes.

7. THE JTPA DATA

The National JTPA (Job Training Partnership Act) Experiment was commissioned by the U.S. Department of Labor for the purpose of evaluating the main U.S. government

training programme for disadvantaged workers. The JTPA programme provides on-the-job training, job search assistance, and classroom training to youth and adults, who qualify for the programme under title IIA of the National Job Training Partnership Act. Persons become eligible for the programme by having a family income near or below the poverty level for six months prior to application or by participating in federal, state or local welfare and foodstamp programmes.¹³ The fact that eligibility is based on a relatively short earnings history makes it possible for some highly skilled or trained individuals to become eligible after a short period of unemployment.

Devine and Heckman (1996) present a detailed analysis of JTPA eligibility rules and of their implications for the eligible population. Barnow (1993) compares the eligibility rules of JTPA to those of other training programmes, including its predecessors, CETA (Comprehensive Employment and Training Act) and MDTA (Manpower Development and Training Act), and finds only minor differences. The JTPA programme we analyse is typical of a wide array of training programmes implemented in the U.S. and abroad.

Under our supervision, the JTPA experiment collected longitudinal data on a group of treatments and randomized-out controls as well as on a comparison group of eligible nonparticipants (ENPs). Two-thirds of programme applicants were assigned to treatment and one third were randomized out and denied access to JTPA services for 18 months to form a control group. Persons were assigned to the control group only after they had applied to the JTPA programme, been declared eligible, and been accepted into the programme. The samples used in this study come from four of the sixteen JTPA training sites participating in the experiment.¹⁴

Members of the eligible nonparticipant comparison group (ENPs) reside in the same narrowly-defined geographic regions as the programme applicants and are eligible for the programme but do not apply to it. ENP comparison group members were administered the same survey instrument as randomized-out controls, which includes detailed retrospective questions on labour force participation, job spells, earnings, marital status, training and schooling activities, transfer programme participation and other demographic characteristics. We combine data from a baseline survey and from two follow-up surveys to form a thirty-six month panel data set.¹⁵ It is divided into the eighteen month period before

13. The specific eligibility criteria are as follows. A person is considered economically disadvantaged and therefore eligible for employment and training services provided under the Act, if he or she: (1) receives, or is a member of a family that receives, cash payments under a Federal, State or local welfare programme; (2) has, or is a member of a family which has, received a total family income for the six month period prior to application for the programme involved (inclusive of unemployment compensation, child support payments and welfare payments), which in relation to family size, was not in excess of the higher of (a) the Office of Management and Budget poverty level or (b) 70% of the lower living standard income level; (3) is receiving food stamps pursuant to the Food Stamp Act of 1977; (4) qualifies as a homeless individual under Section 103 of the Stewart B. McKinney Homeless Assistance Act; (5) is a foster child on behalf of whom State or local government payments are made; or (6) is an adult handicapped individual whose own income meets the requirements of the clause but who is a member of a family whose income does not meet such requirements. (See Job Training Partnership Act of 1982, Public Law 97-300, 29 USC 103.) In addition, JTPA training sites may admit up to 10% of those served under Title IIA for reasons other than eligibility through these criteria. See Devine and Heckman (1996).

14. Kemple, Doolittle and Wallace (1993) provide detailed descriptions of all 16 experimental sites. Eligible nonparticipant data was only collected at four of the sites: Fort Wayne, Indiana; Corpus Christi, Texas; Jersey City, New Jersey; and Providence, Rhode Island.

15. The baseline survey collected retrospective monthly data on demographic characteristics, earnings histories, labour market histories, participation in government transfer programmes and participation in schooling or training activities. A follow-up survey, administered twelve to twenty four months later, collected similar information. The response rate for this survey was around 84%. The sample used includes persons who (1) had a follow-up interview scheduled at least 18 months after random assignment, (2) responded to the survey, and (3) had usable earnings information for at least 14 of the 18 months after random assignment. Appendix E (available on request) and Smith (1994) contain additional information on the design and collection of the ENP sample.